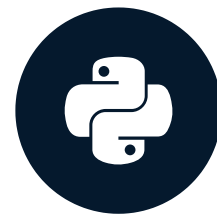


Joining data: a real-world necessity

PANDAS JOINS FOR SPREADSHEET USERS



John Miller
Principal Data Scientist

Pandas for spreadsheet users

- Learn based on similarities to spreadsheets
- Understand the power and flexibility of pandas
- Use data from the National Football League (NFL)



A dense collage of various business and data-related icons. The central element is a large magnifying glass focusing on a pie chart. Surrounding it are numerous other icons: a bar chart, a line graph, a calculator, a laptop displaying a chart, a location pin, a cloud, a percentage sign, a magnifying glass, a bar chart, a pie chart, a line graph, a checkmark, a plus sign, a minus sign, a document, a magnifying glass, a bar chart, a pie chart, a line graph, a checkmark, a plus sign, a minus sign, a document. The icons are in various colors (blue, red, grey, black) and styles (flat, 3D, line art).

- Datasets split by time or other factor
- Datasets with related factors

Split data

- Influenced by reporting cycle
- Common splits
 - Time
 - Geography
 - Business unit

Split data example

	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	1	2016	Pre	Indianapolis Colts
3	2	2016	Pre	Los Angeles Rams
4	3	2016	Pre	Baltimore Ravens
5	4	2016	Pre	Green Bay Packers
6	5	2016	Pre	Atlanta Falcons
games_2016				
Ready				
	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	334	2017	Pre	Dallas Cowboys
3	335	2017	Pre	Arizona Cardinals
4	336	2017	Pre	Baltimore Ravens
5	337	2017	Pre	Buffalo Bills
6	338	2017	Pre	San Francisco 49ers
games_2017				

Split data example

	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	1	2016	Pre	Indianapolis Colts
3	2	2016	Pre	Los Angeles Rams
4	3	2016	Pre	Baltimore Ravens
5	4	2016	Pre	Green Bay Packers
6	5	2016	Pre	Atlanta Falcons
games_2016				
Ready				
	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	334	2017	Pre	Dallas Cowboys
3	335	2017	Pre	Arizona Cardinals
4	336	2017	Pre	Baltimore Ravens
5	337	2017	Pre	Buffalo Bills
6	338	2017	Pre	San Francisco 49ers
games_2017				

Split data example

The image shows a screenshot of a spreadsheet application with two tables. The top table, 'games_2016', has columns: GameKey, Season, SeasonType, and HomeTeam. The bottom table, 'games_2017', has the same columns. A red box highlights the 'GameKey' column in both tables, indicating a join operation.

	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	1	2016	Pre	Indianapolis Colts
3	2	2016	Pre	Los Angeles Rams
4	3	2016	Pre	Baltimore Ravens
5	4	2016	Pre	Green Bay Packers

	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	334	2017	Pre	Dallas Cowboys
3	335	2017	Pre	Arizona Cardinals
4	336	2017	Pre	Baltimore Ravens
5	337	2017	Pre	Buffalo Bills

Complementary data

- Results from collecting data for different purposes
- Department-specific data
- Storage in separate files or database tables

Complementary data example

	A	B	C
1	GameKey	Season	SeasonType
2	1	2016	Pre
3	2	2016	Pre
4	3	2016	Pre
5	4	2016	Pre

	A	B	C
1	GameKey	Turf	Stadium
2	1	Turf	Tom Benson Hall of F
3	2	Grass	Los Angeles Memoria
4	3	Natural Grass	M&T Bank Stadium
5	4	DD GrassMaster	Lambeau Field

Complementary data example

	A	B	C
1	GameKey	Season	SeasonType
2	1	2016	Pre
3	2	2016	Pre
4	3	2016	Pre
5	4	2016	Pre

	A	B	C
1	GameKey	Turf	Stadium
2	1	Turf	Tom Benson Hall of F
3	2	Grass	Los Angeles Memoria
4	3	Natural Grass	M&T Bank Stadium
5	4	DD GrassMaster	Lambeau Field

Complementary data example

	A	B	C
1	GameKey	Season	SeasonType
2	1	2016	Pre
3	2	2016	Pre
4	3	2016	Pre
5	4	2016	Pre

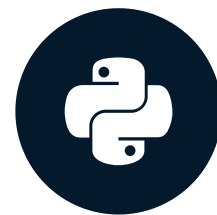
	A	B	C
1	GameKey	Turf	Stadium
2	1	Turf	Tom Benson Hall of F
3	2	Grass	Los Angeles Memoria
4	3	Natural Grass	M&T Bank Stadium
5	4	DD GrassMaster	Lambeau Field

Let's practice!

PANDAS JOINS FOR SPREADSHEET USERS

Concatenation

PANDAS JOINS FOR SPREADSHEET USERS

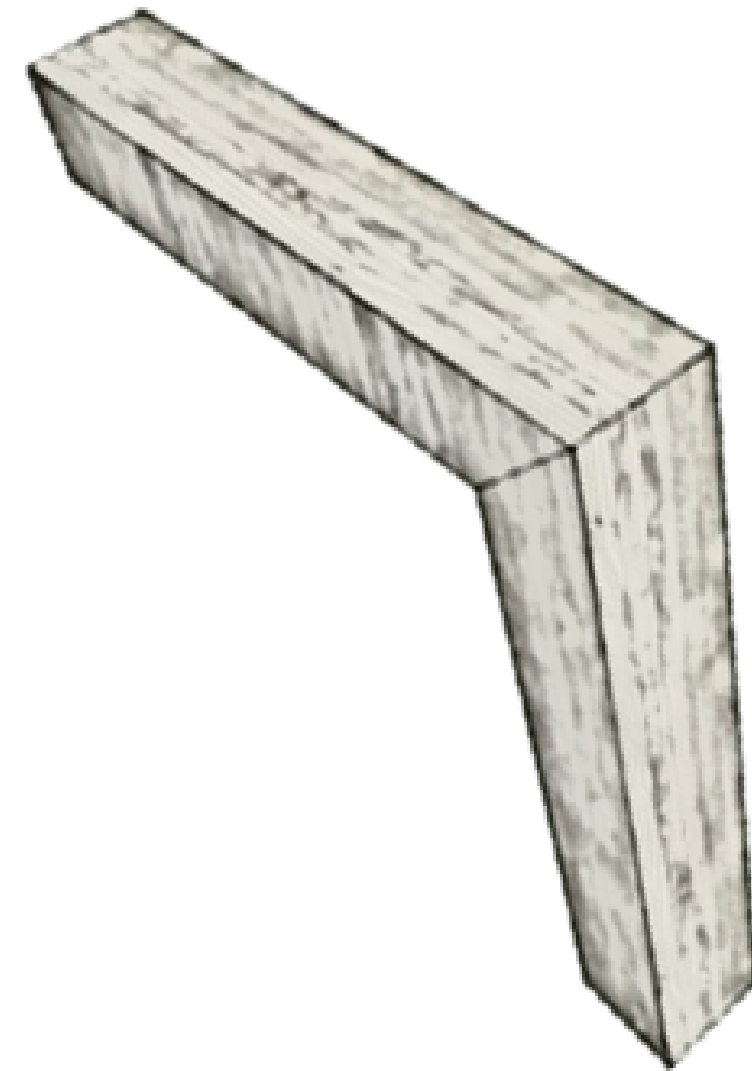


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Concatenation basics

- Similar to spreadsheet CONCATENATE
- Mimics copy-paste of cells
- `pd.concat()` along rows or columns

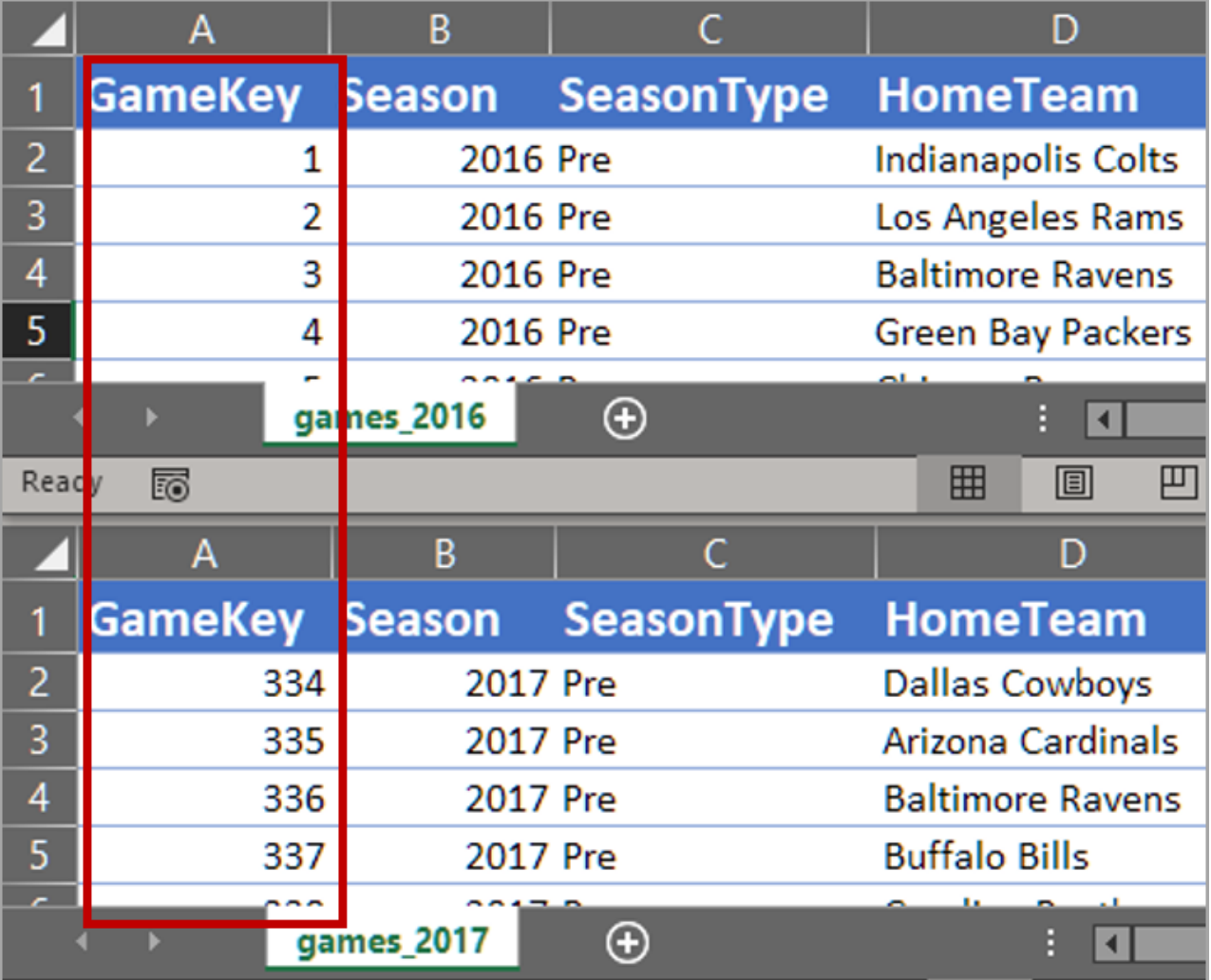


Concatenating rows

- Useful when working with split data

```
pd.concat([df1, df2, ...])
```

- Uses unique key(s) as data frame index
- Includes all rows by default



	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	1	2016	Pre	Indianapolis Colts
3	2	2016	Pre	Los Angeles Rams
4	3	2016	Pre	Baltimore Ravens
5	4	2016	Pre	Green Bay Packers
6	5	2016	Pre	Atlanta Falcons

games_2016

	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	334	2017	Pre	Dallas Cowboys
3	335	2017	Pre	Arizona Cardinals
4	336	2017	Pre	Baltimore Ravens
5	337	2017	Pre	Buffalo Bills
6	338	2017	Pre	San Francisco 49ers

games_2017

Concatenating rows with overlapping indices

- Data frame indices may overlap
- Don't worry!

```
pd.concat([df1, df2, ...],  
          ignore_index=True)
```

The image shows two overlapping data frames in a spreadsheet-like interface. The top frame is labeled 'games_2016' and the bottom frame is labeled 'games_2017'. Both frames have columns A, B, C, and D. The top frame's data is as follows:

	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	1	2016 Pre		Indianapolis Colts
3	2	2016 Pre		Los Angeles Rams
4	3	2016 Pre		Baltimore Ravens
5	4	2016 Pre		Green Bay Packers

The bottom frame's data is as follows:

	A	B	C	D
1	GameKey	Season	SeasonType	HomeTeam
2	334	2017 Pre		Dallas Cowboys
3	335	2017 Pre		Arizona Cardinals
4	336	2017 Pre		Baltimore Ravens
5	337	2017 Pre		Buffalo Bills

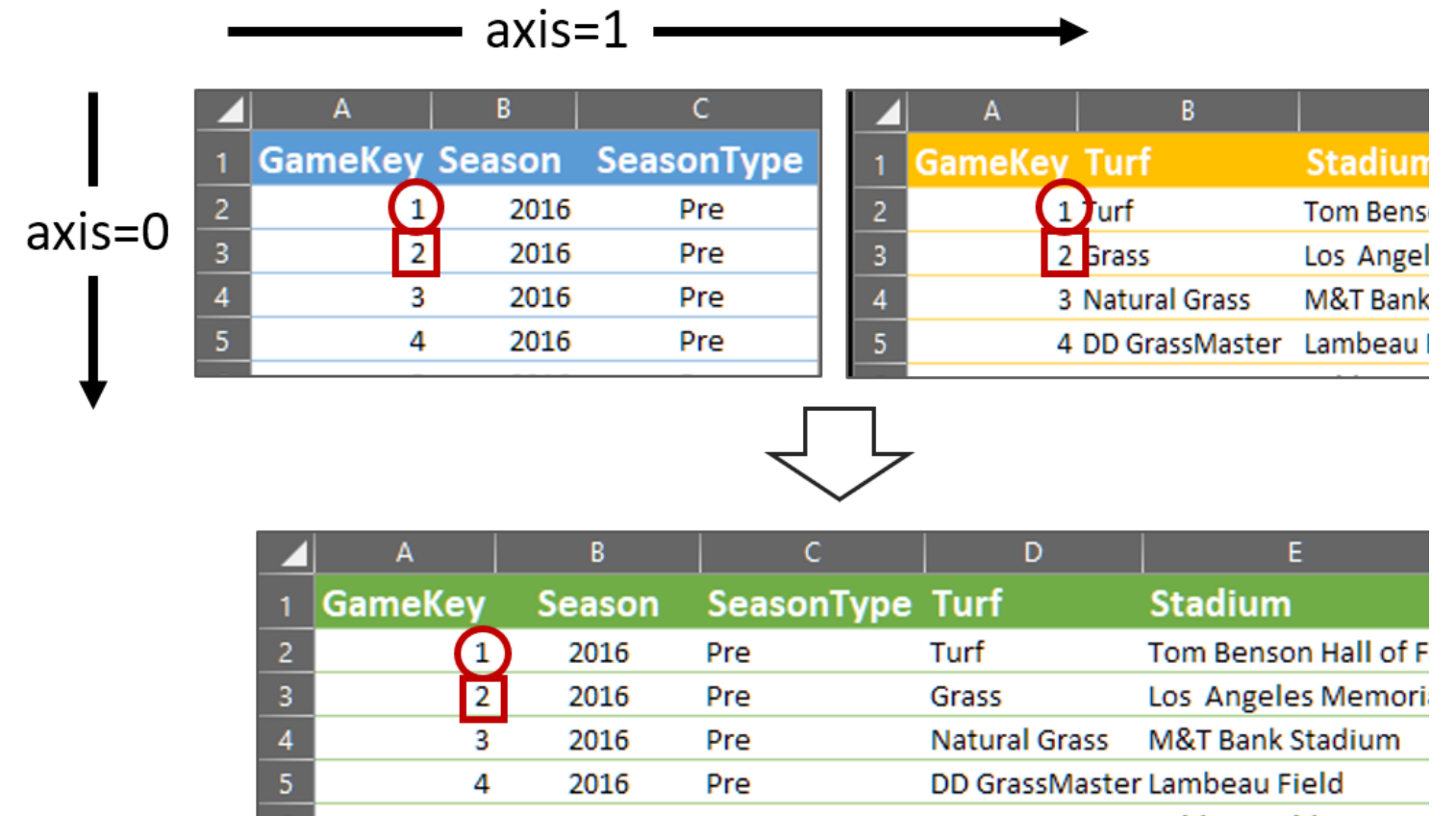
Red circles highlight the overlapping indices (1-5) in both frames.

Concatenating columns

- Like pasting tables side by side
- Across columns: axis=1

```
pd.concat([df1, df2, ...],  
          axis=1)
```

- Includes all columns by default

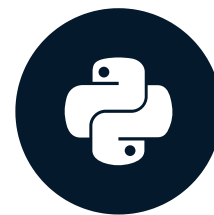


Let's practice!

PANDAS JOINS FOR SPREADSHEET USERS

Power and flexibility

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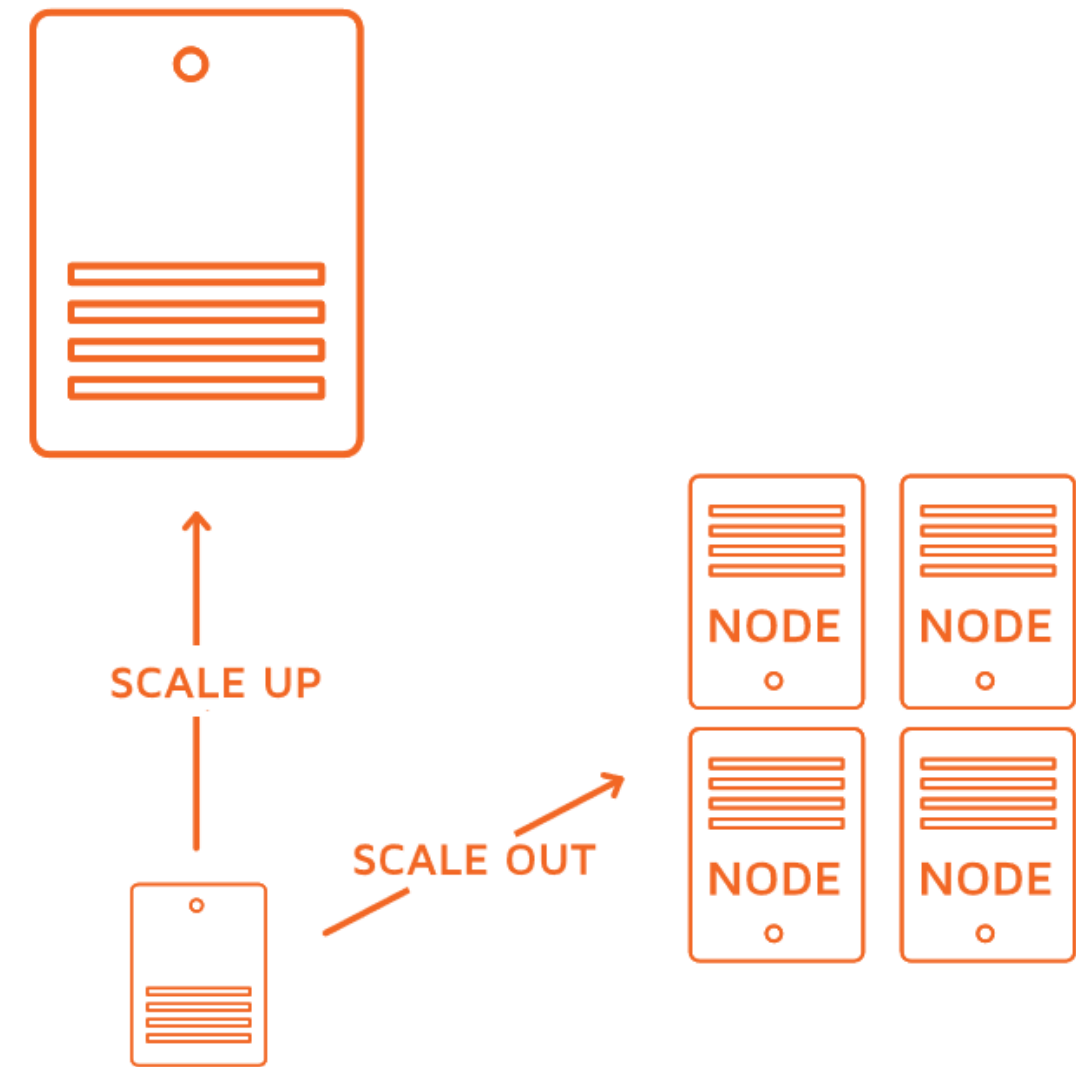


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Scalability

- No hard limits on data frame size
- Built-in ways to "chunk" data
- Use distributed/parallel computing



Efficiency

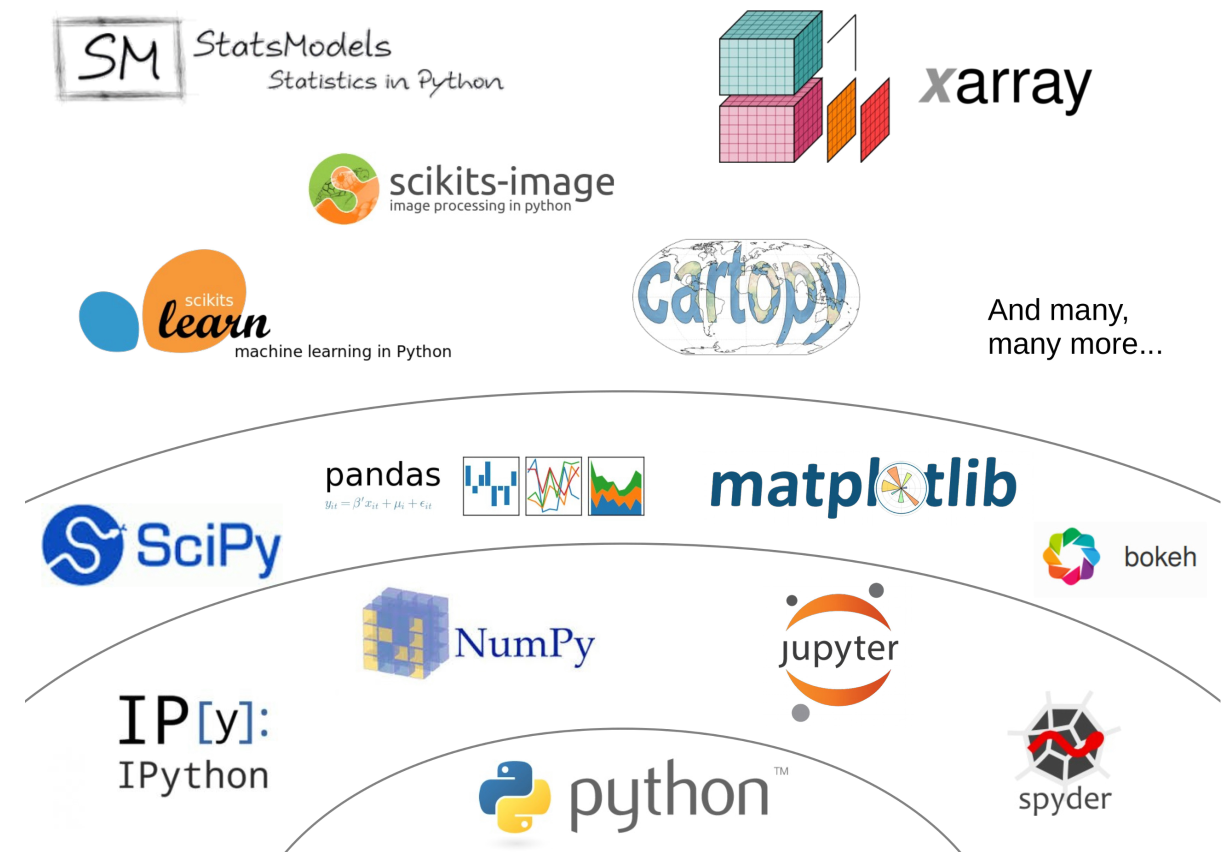


- Join on multiple columns
- Preference for simple code

```
joined_df = left_df.merge(right_df)
```

Integration

- Improved speed and scale
- Data visualization
- Machine learning



A word on advanced spreadsheet usage

- Data models and query tools
- Programming languages
- Advanced formulas

Let's practice!

PANDAS JOINS FOR SPREADSHEET USERS